

Fermat's Little Theorem

1 What It Says

Fermat's Little Theorem

If p is prime and $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

The condition $p \nmid a$ means that a is not divisible by p . Since p is prime, this is the same as

$$\gcd(a, p) = 1.$$

Why It Matters

Fermat's Little Theorem helps reduce very large powers modulo a prime.

$$a^{p-1} \equiv 1 \pmod{p}$$

lets us replace exponents by their remainders modulo $p - 1$.

2 Needed Modular Arithmetic

Congruence

$$a \equiv b \pmod{n} \iff n \mid (a - b).$$

This means a and b leave the same remainder when divided by n .

For example,

$$35 \equiv 2 \pmod{3},$$

because

$$35 - 2 = 33$$

and

$$3 \mid 33.$$

Cancellation Rule

From

$$ac \equiv bc \pmod{n},$$

we can cancel c only when

$$\gcd(c, n) = 1.$$

For Fermat's theorem, the important fact is:

$$\phi(p) = p - 1 \quad \text{when } p \text{ is prime.}$$

3 Proof

Use Euler's Theorem

Euler's theorem says:

$$\gcd(a, n) = 1 \Rightarrow a^{\phi(n)} \equiv 1 \pmod{n}.$$

If p is prime and $p \nmid a$, then

$$\gcd(a, p) = 1.$$

Apply Euler's theorem with

$$n = p.$$

Then

$$a^{\phi(p)} \equiv 1 \pmod{p}.$$

Since p is prime,

$$\phi(p) = p - 1.$$

Therefore,

$$a^{p-1} \equiv 1 \pmod{p}.$$

4 Equivalent Form

Another Common Version

For every integer a and prime p ,

$$a^p \equiv a \pmod{p}.$$

Why?

If $p \mid a$, then both sides are 0 modulo p .

If $p \nmid a$, then Fermat's theorem gives

$$a^{p-1} \equiv 1 \pmod{p}.$$

Multiply both sides by a :

$$a^p \equiv a \pmod{p}.$$

5 Worked Example

Problem

Verify Fermat's theorem for

$$a = 11, \quad p = 13.$$

Since 13 is prime and $13 \nmid 11$, Fermat's theorem says

$$11^{12} \equiv 1 \pmod{13}.$$

Compute using repeated squaring:

$$11^2 = 121 \equiv 4 \pmod{13}.$$

$$11^4 \equiv 4^2 = 16 \equiv 3 \pmod{13}.$$

$$11^8 \equiv 3^2 = 9 \pmod{13}.$$

Since

$$12 = 8 + 4,$$

we get

$$11^{12} = 11^8 \cdot 11^4.$$

Therefore

$$11^{12} \equiv 9 \cdot 3 = 27 \equiv 1 \pmod{13}.$$

Answer

$$11^{12} \equiv 1 \pmod{13}.$$

6 Large Power Example

Problem

Find

$$7^{100} \pmod{13}.$$

Since 13 is prime and $13 \nmid 7$,

$$7^{12} \equiv 1 \pmod{13}.$$

Reduce the exponent modulo 12:

$$100 = 12(8) + 4.$$

So

$$7^{100} = 7^{12 \cdot 8 + 4} = (7^{12})^8 \cdot 7^4.$$

Modulo 13:

$$7^{100} \equiv 1^8 \cdot 7^4 \equiv 7^4 \pmod{13}.$$

Now compute

$$7^2 = 49 \equiv 10 \pmod{13}.$$

Then

$$7^4 \equiv 10^2 = 100 \equiv 9 \pmod{13}.$$

Answer

$$7^{100} \equiv 9 \pmod{13}.$$

7 Final Review

Core Facts

p prime and $p \nmid a \Rightarrow a^{p-1} \equiv 1 \pmod{p}$.

Equivalent form: $a^p \equiv a \pmod{p}$.

Use it to reduce powers modulo a prime.

It follows from Euler's theorem since $\phi(p) = p - 1$.